Week - 4

12/10/2026

Arrays:-

(Homogenous)

1) The arrays is a collection of the same type of elements at contiguos memory.
location under the same name.

2) It has a finite data.
3) It is a easier to access the element in the case of an array.
4) Inertion and deletion operations are costly in the case of an array.
5) ~~e~~ Non. Modification is possible at the runtime after the array is created.
6) Arrays are static.

list:-

1) list use dynamic and can change in size during runtime.
2) list can accomodate elements of different data type. (Non-Homogenous)

★ Not :- It inden sequence is not proper like from 0 to 1, 2, 3, 4.... it will give you a inden error.

eg:- labels = ["R-001", "R-002",
-3 -2
"R-003"]
-1

Slicing :- same rule as Range (start, stop)

— (stop) is never included

example :- values = [10, 20, 30, 40, 50]
0 1 2 3 5
print (values [1:3]

output => (20, 30)

Append = you can add any element at the last of the list.

eg :- x = [1, 2, 3] print (x)
x. append (20) output => [1, 2, 3, 20]

Tuple :- A tuple in python is a collection of items that is ordered and cannot be changed after it is created (this is called immutable).

*Note: Tuble = ( ) -> round bracket
list = [ ] -> square bracket

*Note: list can be change able (mutuabl
Tuple are not (immutable)

Note:- Dictionaries { } -> curly brackets
It stores { — : — , — : — }
↓ ↓
key value
pairs

=> A dictionaries is a date structure that stores information in key - value pairs
eg:- record = { "label" : "Temperature" }
print (record ["label"])
output => Temperature.

#Set:- A set is a collection of unique values. It automatically removes duplicate values and does not maintain a gurantee order. It is mutable.
↓
remove or add element after creating.

#Note:- String " " → Immutable.